



«Approved»
 by the Dean
 Syzdykova Z.A.
 « » of 2024

Syllabus Academic Year 2024-2025

1. General Information	
Course title	Calculus 4
Degree cycle (level)/ major	Bachelor, Group of Mathematical and Computational Science
Year, term	Year 2, Term 1
Number of credits	5
Language of delivery:	English
Prerequisites	Calculus 1, Calculus 2, Calculus 3, Linear algebra
Postrequisites	Numerical Methods
Lecturer(s)	Nurlybekuly Turdybek, PhD., professor, Astana IT University. E-mail: t.nurlybekuly@astanait.edu.kz . Office: Expo, C1 block, 2nd floor C1.3.249
2. Goals, Objectives, and Learning Outcomes of the Course	
1. Course Description	Functions of Complex Variable and Integral Transforms is a basic course for students of science and engineering. The aim of the course is to teach students the principal techniques and methods of analytic function theory. With a long history and rich content, the theory of complex functions is one of the most fundamental branches of mathematics. It has wide and significant applications to other fields not only in mathematics but also in other engineering subjects, like Analytic Number Theory, Mathematical Physics Equations, Fourier Analysis, Mechanics, Signal Processing, etc. Complex analysis has become an indispensable and standard tool of the working mathematicians, physicists, and engineers. Neglect of it can prove to be a severe handicap in most areas of research and application involving mathematical ideas and techniques.
2. Course Goal(s)	The course aims to extend the methods of calculus of multivariable to calculus for complex valued functions. To give elementary concepts of Harmonic analysis.
3. Course Objectives:	Course objectives include: - To demonstrate knowledge of mathematical knowledge. - To understand basic mathematical principles (proving, solving). - To understand parametrization, differentiation, and integration of complex valued functions. - To understand Analytic Functions. To understand Residues and Conformal Mappings. - To understand Laplace and inverse Laplace transforms; properties, convolutions; solution of ODE and PDE by Laplace transform. - To understand Fourier integrals; Fourier transforms, sine and cosine transforms; solution of PDE by Fourier transform - To develop mathematical abilities in writing programs by computers.

4. Skills & Competences	- Ability to write mathematical statements and problem solutions using mathematical symbols. - Understanding of key mathematical concepts and the application of integral transforms.
5. Course Learning Outcomes:	At the end of the course the student is expected to be able to analyse: 1. Complex Analysis: Complex Numbers, Complex Plane, Polar Form of Complex Numbers. Powers and Roots, Derivative. Analytic Functions, Cauchy – Riemann Equations, Laplace Equation, Conformal mapping, Exponential Function, Trigonometric Functions, Hyperbolic Functions, Logarithms, General Power, Linear Fractional Transformation. Complex Line Integral, Cauchy Integral Theorem, Cauchy Integral Formula, Derivatives of Analytic Functions. 2. Laplace Transform: Laplace Transforms, Inverse Transforms, Linearity, Shifting, Transforms of Derivatives and Integrals, Differential Equations, Unit Step Function, Second Shifting Theorem, Dirac's Delta Function. Differentiation and Integration of Transforms. Convolution, Integral Equations, Partial Fractions, Differential Equations, Systems of Differential Equations. 3. Fourier Series: Fourier series, Half range Expansions, Parseval's Identity, Fourier Integrals, Fourier integral theorem. Sine and Cosine Integrals. 4. Fourier Transforms: Sine and Cosine Transforms, Properties, Convolution theorem.
6. Methods of Assessment	• Homework • Quiz • Midterm, Pre-Final and Final exams
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic Literature: 1. Lecture presentations. 2. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Edition, McGraw-Hill, 2003. 3. Larry C. Andrews and Bhimson. K. Shivamoggi, The Integral Transforms for Engineers, Spie Press, Washington, 1999. Supplementary literature: 4. R. J. Beerends, H. G. ter Morsche, J. C. van den Berg and E. M. van de Vri, Fourier and Laplace Transforms, Cambridge University Press, New York, 2003
8. Resources	Online journals, articles, papers, books, and internet resources, as well as online emulators and online software for simulation.
9. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric.

In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course.

In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU.

It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more, he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing.

Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective.
- Contributes to moving the discussion and analysis forward.
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Homework / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone’s work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.

Final exam* / Final project:

The offline final exam for the course “Calculus-4” includes eight theoretical and practical tasks for 80 minutes.

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor.

Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;
4. Submitting the same work for credit in two courses without prior consent of both instructors.

Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university.

Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz).

Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.

** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work

2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (Office hours)	SIS (H/W)
1	Complex numbers, topology in C	3	2		1	
2	Functions of one complex variable, limits, continuity, and differentiability	3	2		1	
3	The Cauchy-Riemann equations with consequences.	3	2		1	
4	Analytic and harmonic functions.	3	2		1	
5	Conformal mappings. Elementary functions from C to C	3	2		1	
6	Fourier Integral Representations. Fourier Transform. Properties of the Fourier Transform.	3	2		1	
7	Transforms of More Complicated Functions. The Convolution Integrals of Fourier. Transforms Involving Generalized Functions	3	2		1	
8	The Laplace Transformation. The Transforms of Some Typical Functions. Basic Operational Properties	3	2		1	
9	Transforms of More Complicated Functions. The Inverse Laplace Transform. Complex Inversion Formula	3	2		1	
10	Applications involving Laplace Transforms. Evaluating Integrals. Solutions of Odes. Solutions of PDEs.	3	2		1	
	Total hours: 50	30	20		10	

3.2 List of Assignments for Student Independent Study

Nº	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Reflection principle	9	[2], Chapter 2	Reading list, books, internet resources

				electronic version
2	Liouville's theorem and fundamental theorem of algebra, maximum modulus principle	9	[2], Chapter 4	Reading list, books, internet resources electronic version
3	Laurent series and properties	9	[2], Chapter 5	Reading list, books, internet resources electronic version
4	Residues and poles	9	[2], Chapter 6	Reading list, books, internet resources electronic version
5	The Mellin Transform	9	[3], Chapter 6	Reading list, books, internet resources electronic version
6	Signals and systems	9	[4], Part 1	Reading list, books, internet resources electronic version
7	The Hankel Transform	9	[3], Chapter 7	Reading list, books, internet resources electronic version
8	Applications of the Fourier integral transforms on PDFs.	9	[3], Chapter 3	
9	Applications of the Laplace integral transforms on PDFs.	9	[3], Chapter 5	

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments**: assignment 1, assignment 2 Quiz 1 Mid Term	60 20 20 20 40	100
2 nd attestation	Assignments**: assignment 3, assignment 4, Quiz 2 End Term	60 20 20 20 40	100
Final exam*	Written Exam		100
Total	$0,3 * 1^{st} \text{ Att} + 0,3 * 2^{nd} \text{ Att} + 0,4 * \text{Final}$		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
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A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Nurlybekuly Turdybek	Professor	23/08/2024	

Director



Sergaziyev M.Zh.